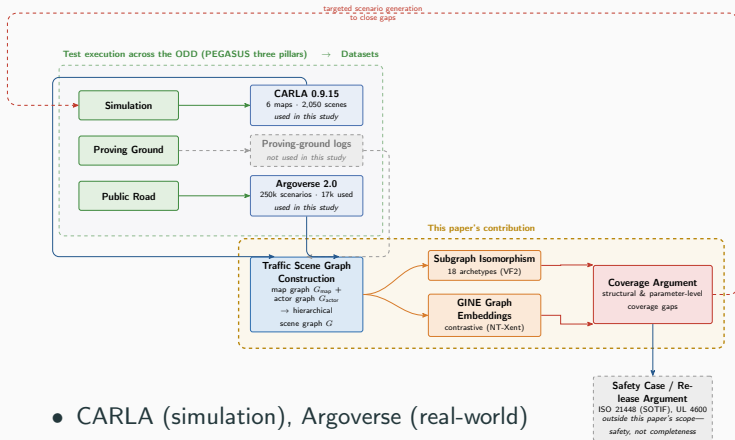


A Graph-based Framework for Coverage Analysis in Autonomous Driving

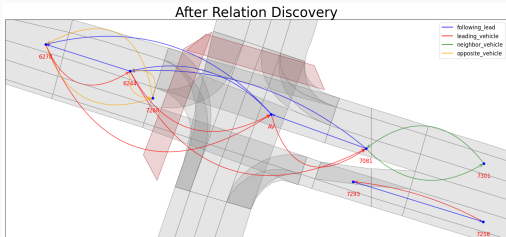
Thomas Mühlenstädt Marius Bause

Verification and Validation Workflow

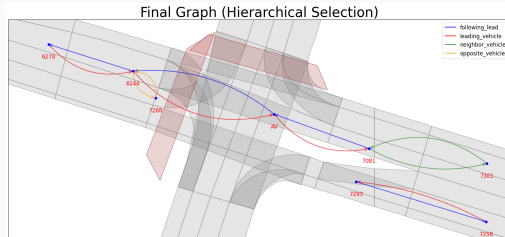


- CARLA (simulation), Argoverse (real-world)
- Ours: coverage argument via graph representation

Hierarchical Scene Graph Construction



After relation discovery

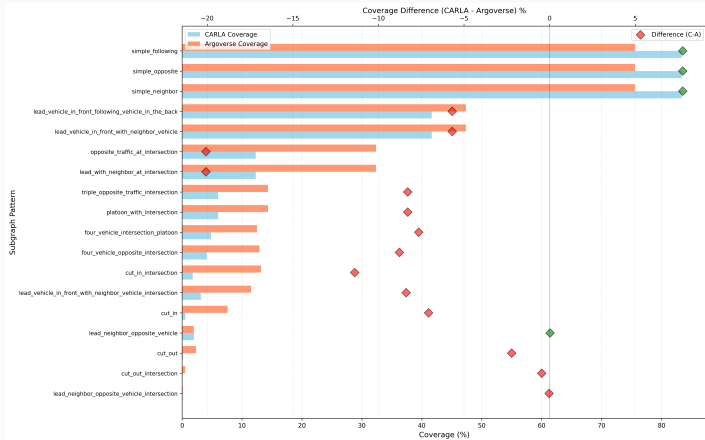


After hierarchical selection

- Map graph edges types included: following, neighbor, opposite
- Lane-change indicator: lane change within $\Delta t = 1s$
- Phase 1 – Relation Discovery → Phase 2 – Hierarchical Construction by selecting by edge type and distance

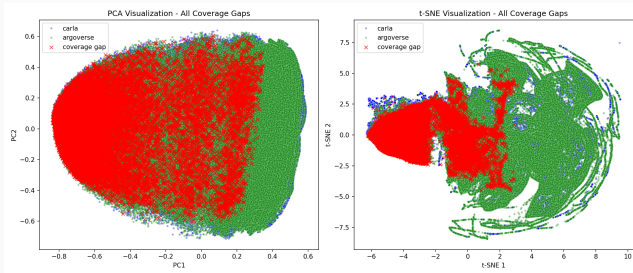
Top-Down Analysis: Scenario Archetype Coverage Gaps

- Archetypes: 18 types user-defined (can be detected by algorithm)
- Subgraph matching via the **VF2** algorithm — near-linear in the number of nodes of the scene graph (sparse graphs)
- Result: missing archetypes reveal coverage gaps



Bottom-Up Analysis: Embedding Space Coverage Gaps

- **GINE**: edge-aware graph embeddings
- Self-supervised contrastive training (CARLA + Argoverse)
- Plausibility check: nearest neighbours match
- First 2 principal components: 40% variance
- 23,148 Argoverse graphs flagged as gaps



Summary and Future Work

- Graph-based framework: hierarchical scene graphs combining map topology with actor relationships
- Two complementary coverage methods:
 - subgraph isomorphism against archetypes → interpretable, pattern-based coverage
 - GINE embeddings via contrastive learning → similarity-based coverage, clustering, anomaly detection
- Validated on Argoverse 2.0 and CARLA: reveals structural, parametric, and co-occurrence coverage gaps in simulation
- Scales across scenarios without scenario-specific handling
- Naturally accommodates varying numbers of actors
- **Future work:**
 - multi-timestep temporal graphs
 - automatically extracting new archetype candidates from dense, unmatched embedding regions
- Code: https://github.com/tmuehlen80/graph_coverage